

A worked example: *orbits* and the labels (μ, ν, c) a triangle with a center dot

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This is the smallest example that shows what an **orbit** is and what the three symmetry labels mean — the *shape* μ , the *teammate* ν , and the *copy* c — with every mode written out explicitly.

1 The object and the group

Object: four grid points — a center dot and three corners of an equilateral triangle:

$$|O\rangle \text{ (center), } \quad |A\rangle, |B\rangle, |C\rangle \text{ (corners).}$$

Group: the 6 symmetries of the triangle (D_3): the identity e , two rotations $r = +120^\circ$, $r^2 = -120^\circ$, and three flips s_A, s_B, s_C (each flip is the mirror through one corner and the midpoint of the opposite side). How they move the points:

- rotations cycle the corners $A \rightarrow B \rightarrow C \rightarrow A$ and leave O fixed;
- flip s_A fixes A and O , and swaps $B \leftrightarrow C$ (similarly s_B, s_C).

2 Orbits — points shuffled among themselves

An **orbit** is a set of points the operations move only among each other. There are two:

$$\textbf{Orbit 1} = \{O\} \text{ (size 1, fixed by everything), } \quad \textbf{Orbit 2} = \{A, B, C\} \text{ (size 3).}$$

3 The modes (symmetry-adapted standing waves)

Inside each orbit we build combinations that behave simply under the group. These are exactly the standing-wave / normal modes.

From Orbit 1 $\{O\}$ — one mode, and it is left unchanged by every operation:

$$|O\rangle.$$

From Orbit 2 $\{A, B, C\}$ — three modes: one “breathing” mode (all corners in phase) and a locked pair of “tilt” modes (corner weights summing to zero):

$$\underbrace{\frac{1}{\sqrt{3}}(|A\rangle + |B\rangle + |C\rangle)}_{\text{breathing}}, \quad \underbrace{\frac{1}{\sqrt{6}}(2|A\rangle - |B\rangle - |C\rangle)}_{\text{tilt 1}}, \quad \underbrace{\frac{1}{\sqrt{2}}(|B\rangle - |C\rangle)}_{\text{tilt 2}}.$$

Four points in, four modes out.

4 The labels (μ, ν, c)

- $\mu = \mathbf{shape}$ (symmetry type): here *symmetric* (unchanged by every op) or *E* (the tilt pair). The third possible shape of D_3 , the *sign* shape, does not occur here.
- $\nu = \mathbf{teammate}$: some shapes come as a locked group. *E* is a pair, so $\nu = 1, 2$; the symmetric shape is a team of one.
- $c = \mathbf{copy}$: if the same shape shows up more than once, number the repeats.

μ (shape)	ν (teammate)	c (copy)	mode
symmetric	1	1	$ O\rangle$
symmetric	1	2	$\frac{1}{\sqrt{3}}(A\rangle + B\rangle + C\rangle)$
<i>E</i>	1	1	$\frac{1}{\sqrt{6}}(2 A\rangle - B\rangle - C\rangle)$
<i>E</i>	2	1	$\frac{1}{\sqrt{2}}(B\rangle - C\rangle)$

Count check: symmetric (2 copies $\times$ team-size 1) + *E* (1 copy $\times$ team-size 2) = 2 + 2 = 4 modes = 4 points. (Multiplicities from characters: symmetric $\times 2$, *E* $\times 1$, sign $\times 0$.)

5 Watch the group act — this is what the labels mean

Symmetric (μ): every operation leaves $|O\rangle$ and $\frac{1}{\sqrt{3}}(|A\rangle + |B\rangle + |C\rangle)$ *exactly unchanged*. That invariance *is* what “symmetric shape” means.

E teammates (ν): the rotation r turns the tilt pair into *each other* — it acts on (tilt 1, tilt 2) as a genuine 120° rotation,

$$r : \begin{pmatrix} \text{tilt 1} \\ \text{tilt 2} \end{pmatrix} \mapsto \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} \text{tilt 1} \\ \text{tilt 2} \end{pmatrix}.$$

Neither tilt is left alone; they only make sense as a pair. That is why *E* has two teammates ($\nu = 1, 2$) and cannot be split — exactly like the coordinates (x, y) under rotation.

Copies (c): $|O\rangle$ and $\frac{1}{\sqrt{3}}(|A\rangle + |B\rangle + |C\rangle)$ are *two independent* symmetric modes (one on the center, one on the corners). Same shape, different states $\Rightarrow$ two copies.

6 What a symmetry-respecting operator looks like

Any operator W that commutes with all 6 operations is block-diagonal by shape, acts as identity on teammates, and mixes only copies:

$$W = \underbrace{\begin{pmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{pmatrix}}_{\text{symmetric block, on } c=1,2} \oplus \underbrace{w_E \cdot \mathbb{I}_2}_{\text{E block, same on both teammates}}.$$

So W may **mix** $|O\rangle$ **with the corner-sum** (both symmetric), and merely **scales the tilt pair** (identical factor on $\nu = 1, 2$). It never changes a shape and never mixes teammates. The only free numbers are $\sum_\mu n_\mu^2 = 2^2 + 1^2 = 5$ — a 2×2 block plus a 1×1 — even though W is a 4×4 operator.

7 Dictionary

term	plain meaning
orbit	a set of grid points the operations shuffle only among each other
μ (shape)	the symmetry type of a mode; the group's fixed "menu" of shapes
ν (teammate)	which member of a locked, symmetry-degenerate group of modes
c (copy)	which repeat, when the same shape occurs more than once (pooled across orbits)

Map to the crystal problem. grid points $\leftrightarrow$ the ISDF interpolation grid; group $\leftrightarrow$ the space group; shape $\mu \leftrightarrow$ (star, little-group irrep); teammate $\nu \leftrightarrow$ (star member, little partner); copy $c \leftrightarrow$ multiplicity (the material data); $W \leftrightarrow$ the Coulomb tensor, whose only free content lives in the copy blocks w_i . The shapes and the way teammates rotate are fixed by the group (material-free); the copies and the block matrices w_i are the material's.

8 Within-orbit multiplicity: one orbit repeating a shape

Above, no shape repeated *inside* a single orbit — each orbit made each shape at most once, and the *two* symmetric copies came from two *different* orbits (center vs. corners). The smallest case where a *single* orbit repeats a shape needs a **generic point**.

Setup. Place a dot at a *generic* spot inside the triangle — *not* on any mirror line. Applying the 6 operations spreads it to **6 points**, and they form **one orbit** (all six are shuffled among each other). Geometrically they are two mirror-image triangles,

$$\text{triangle 1: } A, B, C, \quad \text{triangle 2: } A', B', C',$$

glued into a single orbit because a reflection swaps the two triangles.

Modes. The 6 modes split as:

- *symmetric* (A_1), 1 mode: $\frac{1}{\sqrt{6}}(|A\rangle + |B\rangle + |C\rangle + |A'\rangle + |B'\rangle + |C'\rangle)$ — all six in phase;
- *sign* (A_2), 1 mode: $\frac{1}{\sqrt{6}}(|A\rangle + |B\rangle + |C\rangle - |A'\rangle - |B'\rangle - |C'\rangle)$ — the two triangles out of phase (it flips sign under a reflection);
- *E*, **twice** (4 states): the tilt patterns of the two triangles together span a 4-dimensional space carrying *E* *two* times — two copies, each a 2-member pair. (Which combinations you call copy 1 vs. copy 2 is the copy-mixing gauge freedom.)

Count: $1 + 1 + 2 \cdot 2 = 6 =$ orbit size. So this *single* orbit produces the *E* shape **twice** — $c = 1, 2$ both from *one* orbit. That is within-orbit multiplicity.

Why it repeats here but not for the corners. The corners (size-3 orbit) sit *on* the mirror lines — each corner is fixed by a reflection (a nontrivial stabilizer), and that stabilizer “uses up” one *E* copy, leaving *E* once. The generic dot (size-6 orbit) sits on *no* mirror — trivial stabilizer, a *free* orbit (the regular representation), which fits the full *E* twice.

Rule. The more generic the point (smaller stabilizer, bigger orbit), the more copies of each shape fit inside its one orbit; a free orbit fits d_μ copies of shape μ , so *every shape of dimension ≥ 2 repeats*.

Why 6 is the smallest such example. Repeating a shape needs a shape of dimension ≥ 2 (a locked pair), which requires a *nonabelian* group; the smallest is the triangle's (order 6), whose free orbit is 6 points. Abelian groups have only 1-dimensional shapes, so no orbit ever repeats a shape.

Two sources of copies, restated. The global multiplicity $n_\mu = \sum_o a_\mu^o$ pools copies from (i) the same shape repeating *within* one (free/generic) orbit — this section — and (ii) the same shape appearing *across different* orbits — the center-vs-corners case in §5. In the 4-point example only (ii) was active; here only (i).